

2. Preliminaries

In this section, we briefly recall main definitions, notations of some basics of q -calculus that will be useful to us in this paper. These notions are presented in [1, 3, 6] among many others.

Throughout this paper we suppose $0 < q < 1$. The q -analog $[n]_q$ of any positive integer $n \in \mathbb{N}^*$ and q -analog factorial $[n]_q!$ are defined as

$$[n]_q = \frac{1 - q^n}{1 - q} \quad \text{and} \quad [n]_q! = [n]_q \times [n-1]_q \times \cdots \times [1]_q.$$

The q -analog of $(x - a)^n$ is

$$(x - a)_q^n = \begin{cases} 1 & \text{if } n = 0 \\ (x - a) \times (x - qa) \times \cdots \times (x - q^{n-1}a) & \text{if } n \geq 1, \end{cases}$$

and we have

$$(a - x)_q^n = (-1)^n q^{\frac{n(n-1)}{2}} (x - q^{1-n}a)_q^n.$$

The q -analog of the exponential function e^x given by

$$e_q^x = \sum_{k=0}^{+\infty} \frac{x^k}{[k]_q!}.$$

The q -analog of identity $e^x e^{-x} = 1$ is defined by $e_q^x E_q^{-x} = 1$, where

$$E_q^x = e_{\frac{1}{q}}^x = \sum_{k=0}^{+\infty} q^{\frac{k(k-1)}{2}} \frac{x^k}{[k]_q!}.$$

The q -analog of $e^{-\frac{x^2}{2}}$ is given by

$$E_{q^2}^{-\frac{q^2 x^2}{[2]_q}} = \sum_{k=0}^{+\infty} \frac{q^{k(k+1)} (q-1)^k}{(1 - q^2)_{q^2}^k} x^{2k}.$$

The q -analog of an improper integral is proper integral with limits $-\nu$ and ν where

$$\nu = \nu(q) = \frac{1}{\sqrt{1-q}}.$$

For $a, b \in \mathbb{R}$, the Jackson integral or q -integral of arbitrary function $f : \mathbb{R} \rightarrow \mathbb{R}$ on $[a, b]$ is defined by

$$\int_a^b f(x) d_q x = (1 - q) \sum_{k=0}^{+\infty} q^k [b f(q^k b) - a f(q^k a)], \quad (2.1)$$

When f is continue on $[a, b]$, we get that

$$\lim_{q \rightarrow 1} \int_a^b f(x) d_q x = \int_a^b f(x) dx.$$